INCH-POUND
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PERFORMANCE SPECIFICATION

REGULATOR, ENGINE GENERATOR: SOLID STATE 28 VOLT, DUAL CURRENT LIMIT, DIRECT CURRENT

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid state regulator utilizing electronic devices, to regulate the voltage, control the reverse current, and limit the current of the 28 volt (V), 300 ampere (A), direct current (dc) generator.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2920

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-G-62061	- Generator, Engine Accessory: 28 Volt DC, Rated Output 300 Ampere.
MIL-B-62346	- Batteries, Storage: Lead-Acid, Low Maintenance (Metric).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

ARMY

12333040 - DC Solid State Voltage Regulator.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D522 - Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings (DoD Adopted).

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshochocken, PA 19428-2959.)

GENERAL MOTORS ENGINEERING STANDARDS

GM9540P - Accelerated Corrosion Test.

(Application for copies of GM publications may be obtained from Global Engineering, 7730 Corondelet Avenue, Suite 407, St. Louis, MO 63105.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.2 Materials. Materials shall be as specified herein, in applicable standards and specifications, and on applicable drawings. Materials not specifically designated shall be suitable for use in the regulator operating over an ambient temperature range -65 to 215 degrees Fahrenheit (°F) (-53 to 101 degrees Celsius (°C)) without any change in physical or dimensional properties while meeting all performance and environmental requirements of this specification (see 4.5.4.1).

3.2.1 Hazardous materials. Asbestos, cadmium, and radioactive material (see 6.3.4) shall not be used in this item.

3.2.1.1 Radioactive material. Radioactive material is defined by Title 10, Code of Federal Regulations, Part 40, and other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Environmental conditions.

3.3.1 High temperature. The regulator shall meet the requirements of 3.9.1, 3.9.2, 3.9.5 through 3.9.7, 3.9.9, and 3.9.10, when exposed to temperatures up to 220°F (104°C); the regulator shall meet the requirements of 3.9.2, except the regulated voltage shall be 26.5 $\pm$0.7 V. After exposure, the regulator shall meet the requirements of 3.9.1, 3.9.2, 3.9.4a through d, and 3.9.5 (see 4.6.4.1).

3.3.2 Low temperature. The regulator shall meet the requirements of 3.9.1, 3.9.2, 3.9.5 through 3.9.7, 3.9.9, and 3.9.10, after storage at temperatures down to -80°F (-62°C) and during exposure to temperatures down to -65°F (-54°C). The regulator shall also meet the requirements of 3.9.2, except that the regulated voltage shall be 29.5 $\pm$0.7V. Subsequently, the regulator shall meet the requirements of 3.9.1, 3.9.2, 3.9.4a thru d; and 3.9.5 (see 4.6.4.2).

3.3.3 Vibration. When tested as specified in 4.6.4.5, the regulator shall evidence no breakage, loosening of parts, abrasion, or other physical damage. Subsequent to the test, the regulator shall meet the requirements of 3.9.1, 3.9.2, 3.9.4, and 3.9.5.

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3.3.4 Shock. When tested as specified (see 4.6.4.6), the regulator shall evidence no breakage, loosened or distorted parts, or other physical damage. Subsequent to the test, the regulator shall meet the requirements of 3.9.1, 3.9.2, 3.9.4, and 3.9.5.

3.3.5 Corrosion resistance. The regulator shall have corrosion resistance equal or better than 1020 steel sheet, galvanized with a minimum 0.75 mil zinc coating, zinc phosphate pretreatment, epoxy prime paint and CARC top coat. No servicing, adjustment or replacement parts shall be necessary (see 4.6.4.4).

3.3.6 Fungus resistance. The regulator shall not support microbial growth that would adversely affect performance (see 4.6.4.3).

3.3.7 Waterproofness. The regulator shall meet the waterproofness requirements in accordance with 4.6.4.7. Subsequent to exposure, the regulator shall meet the requirements of 3.9.2.

3.4 Weight. The regulator weight shall be no more than 18.5 pounds (lbs) (8.39 kilograms (kg)).

3.5 Design.

3.5.1 Polarity. The regulator shall be designed to operate with a negative-grounded, electrical system, and shall be internally case grounded (see 4.5.4.1).

3.5.2 Electromagnetic interference suppression. Electromagnetic interference generated within the regulator shall be suppressed within the regulator housing (see 4.5.4.1).

3.5.3 Configuration. The regulator fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline of Drawing 12333040. Electrical connectors and mounting dimensions shall also conform to the detail requirements of this drawing. Contact designation shall be in accordance with figure 1 (see 4.5.4.2).

3.6 Construction and hardware.

3.6.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces (see 4.5.4.1).

3.6.2 Standard parts. Military standard parts shall be used wherever possible (see 4.5.4.1).

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3.7 Reliability. Quantitative reliability numbers are to be determined by Government reliability engineers prior to finalizing the contract (see 6.2).

3.8 Endurance. The temperature stabilized regulator (see 6.3.1) shall be capable of withstanding the endurance load cycles specified herein for a total of 1100 hours. Subsequently, the regulator shall meet the requirements of 3.3.7 and 3.9.1 through 3.9.6 (see 4.6.4.9).

3.9 Performance characteristics.

3.9.1 Reverse current. The reverse current shall not exceed 30 millamperes (mA) when measured between the regulator and the batteries (see 4.6.1.1.1).

3.9.2 Voltage regulation. When pin E of the battery connector is not connected to ground, the regulator shall establish and maintain regulated voltage per figure 8 throughout the generator speed range of 2350 to 8160 revolutions per minute (rpm) with load current from battery only to 350 A. With pin E of the battery connector connected to ground, the regulator shall establish and maintain regulated voltage per figure 8 throughout the generator speed range of 3000 $\pm$50 rpm with load current from battery only to 150 A. The fundamental frequency of the voltage waveform shall be greater than 100 Hertz (Hz). The regulator shall re-establish and maintain a regulated voltage within the time and voltage limits indicated on figure 2. Overshoot and undershoot limits shall conform to limits on figure 2. For load currents above 350 A with load increased to 400 A and with pin E of the battery terminal ungrounded, the regulated voltage shall be equal to or less than that measured at 350 A. For loaded currents above 150 A with pin E of the battery connector grounded, the regulated voltage shall be equal to or less than that at 150 A (see 4.6.1.1.2).

3.9.3 Voltage ripple. The regulator output voltage waveform shall have the following limitations: with batteries disconnected, the alternating current (ac) peak to peak ripple voltage shall be no more than 7 V with pin E of the battery connector both grounded and ungrounded. With batteries connected, the peak to peak ripple voltage shall be no greater than 4 V, except that excursions of less than 1 millisecond (ms) may exceed 4 V (see 4.6.1.1.3).

3.9.4 Current limit. The regulator shall provide current control per figure 9 and as specified below (see 4.6.1.1.4):

a. With pin E of the regulator battery connector grounded, the regulator shall limit the generator output to 195 $\pm$15 A throughout the operating ambient temperature range of -65 to 220°F (-53 to 104°C), and a generator speed of 3000 rpm.
b. With pin E of the regulator battery connector grounded, the regulator-generator system shall produce no less than 150 A output current within the temperature adjusted voltage regulation range at a generator speed of 3000 rpm.

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c. With pin E of the battery connector ungrounded, the regulator shall limit the generator output to 400 ±20 A throughout the operating ambient temperature range of -65 to 220°F, and a generator speed of 5500 rpm.
d. With pin E of the battery connector ungrounded, the regulator-generator system shall produce no less than 350 A within the temperature adjusted voltage regulation range at a generator speed of 5500 rpm.
e. The regulator shall attain stable current regulation in the current limit mode within 0.3 second of load application as shown in figure 3.
f. The regulator shall attain stable voltage regulation upon load release within 0.3 second after entering the lower voltage regulation limit of 27.3 V as shown in figure 3.
g. In the current limit mode, the frequency of oscillation of the dc voltage shall have a period of less than 14 ms (see Δt in figure 3 and a peak-to-peak value) (see Δv in figure 3 of less than 10 percent of the dc voltage.

3.9.5 Overvoltage. The regulator shall incorporate an overvoltage protection scheme to protect the regulator-generator system from any failure in the regulator which will cause the regulator to be turned “full on.” The overvoltage portion of the regulator shall deactivate the generator when the system voltage reaches a value of 33 ±1 V. Overvoltage reaction time shall be not less than 0.25 second and not more than 1 second. This criteria shall be met with pin E of the battery connector both grounded and ungrounded. Reset shall be manual and located on the regulator (see 4.6.1.1.5).

3.9.6 Field excitation. The regulator shall be activated and begin to regulate when the generator residual voltage increases above 0.7 V with pin E of the battery connector both grounded and ungrounded (see 4.6.1.1.6).

3.9.7 Field circuit. The regulator shall be protected against any inductive energy that may be applied as a result of opening switch S1 shown in figure 4. Subsequently, the regulator shall meet the requirements of 3.9.2 (see 4.6.1.1.7).

3.9.8 Negative voltage operation. The regulator shall not be damaged by the application of negative voltage. Subsequent to the application of negative voltage, the regulator shall meet the requirements of 3.9.2, 3.9.4, and 3.9.5 (see 4.6.1.1.8).

3.9.9 Transient voltage. The regulator shall withstand the transient environment conforming to 4.6.1.1.9.

3.9.10 Negative transient voltage. The regulator shall evidence no damage when subjected to negative transient voltage (see 4.6.1.1.10).

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3.9.11 Electromagnetic interference. The regulator shall meet the requirements of CE102 and RE102 of MIL-STD-461 (see 4.6.4.8).

3.9.12 Output circuitry. The regulator output circuitry shall be such that two regulator-generator systems (one having pin E of the battery connector ungrounded, and the other having pin E grounded) can be connected to the same battery bus, and operated either independently or simultaneously without damage to either regulator and without detrimental effects to overall system operation (see 4.5.4.1).

3.9.13 Short circuit protection. Regulator circuitry shall be provided to prevent damage to internal components in the event of a short circuit from pin "D", "F" or "H" of the generator connector to ground during normal regulator operations. This criteria shall be met with pin E of the battery connector both grounded and ungrounded (see 4.6.1.1.11).

3.9.14 Generator accessory sensing circuit. Pin "H" of the generator connector shall be capable of supplying 0.5 A at a voltage within 250 mV of the generator output voltage over the operating temperature range of the regulator (see 4.6.1.1.12).

3.10 Physical characteristics.

3.10.1 Accessibility. Access to internal components of the regulator shall be attained by removal of the cover. Cover seals shall be provided to prevent entry except by breaking the seals. Installation and removal of the regulator cover shall be possible without the use of special tools (see 4.5.4.1).

3.10.2 Mounting position. The regulator shall be designed to operate when mounted in any position and without the use of shock absorbing material or equipment (see 4.5.4.1).

3.11 Finishes. All surfaces shall be cleaned, primed, treated and finished in accordance with manufacturer’s standard commercial practices. Unless otherwise specified (see 6.2), all exterior surfaces shall be gloss white, except for electrical connectors and mating surfaces on the mating bracket (see 4.5.4.2).

3.12 Nameplate. A nameplate shall be attached to the regulator housing in the manner specified on the applicable drawings (see 4.5.4.2). The legend contained on the plate shall include at least the following:

a. Regulator, 28 volts dc
b. Manufacturers identification
c. Serial number
d. Military part number
e. Federal stock number

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f. Contract number
g. Date of manufacture
h. US

3.13 Interchangeability. The regulator and its replacement parts shall be completely interchangeable without modification to mountings, physical dimensions or electrical capacities and functions with other regulators of the same contracted manufacturer or other manufacturers (see 4.5.4.2).

3.14 Workmanship. Workmanship shall be of a quality which shall assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connections, cracked insulation, or other defects which affect serviceability, appearance or performance (see 4.5.4.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspections (see 4.5).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 80 ±15°F (26 ±8°C)
b. Barometric pressure: 28.5 (+2, - 3 inches Hg (10.10 +0.01 Pascal (Pa))
c. Relative humidity: 50 ± 30 percent.

4.2.1 Test circuit. Unless otherwise specified herein, the test circuit shown on figure 4 shall include the regulator mounted on a horizontal plane, batteries, and a generator. The generator used in the test circuit shall generate 28 Vdc with 300 ampere rated output. The batteries shall consist of four 12 V batteries each with a reserve capacity of 200 minutes. The four batteries shall be connected in series-parallel to form a 24 V source. Batteries shall be fully charged (see 6.3.2) and kept at room temperature.

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TABLE I. Measuring instrument tolerances.

Measuring instrument	Accuracy	
	First article	Conformance and control
Voltmeter dc	±0.35 percent (40 V-full scale)	±0.15 V
Ammeter dc	+0.50 percent (600 A-full scale)	±3.0 A
Tachometer	±1 percent	±50 rpm at 1900 rpm ±200 rpm at 10 000 rpm
Milliammeter	±0.50 percent (600 mA - full scale)	±5.0 mA
Oscilloscope	±3 percent	±3 percent

4.3 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance, except as specified in table I.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four regulators produced under the production contract for first article inspection. First article samples shall be inspected as specified in table II, in the order specified in table III. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply regulators that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

NOTE: Regulators subjected to first article inspection shall not be used for any other purpose and shall be indelibly marked: "DO NOT USE."

4.5 Conformance inspection.

4.5.1 Sampling.

4.5.1.2 Sampling for examination. Sampling for conformance examination shall be in accordance with the quality assurance section of the contract.

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TABLE II. Classification of inspection.

Table	Requirement	Inspection	First article	Conformance	
				Examination	Tests (see 4.5.3)
Materials and construction	3.5.1, 3.5.2, 3.6.1, 3.6.2, 3.9.14, 3.10, 3.10.1, 3.10.2	4.5.4.1	X		
Defects (see 4.5.2)	3.5.3, 3.11, 3.12, 3.13, 3.14	4.5.4.2	X	X	
Weight	3.4	4.6.2	X		
Reverse current	3.9.1	4.6.1.1.1	X		
Voltage regulation	3.9.2	4.6.1.1.2.1, 4.6.1.1.2.2	X		X
Voltage ripple	3.9.3	4.6.1.1.3	X		
Current limit	3.9.4	4.6.1.1.4.1, 4.6.1.1.4.2	X		X
Overvoltage	3.9.5	4.6.1.1.5.1, 4.6.1.1.5.2	X		X
Field excitation	3.9.6	4.6.1.1.6	X		X
Field circuit	3.9.7	4.6.1.1.7	X		
Negative voltage operation	3.9.8	4.6.1.1.8.1, 4.6.1.1.8.2	X		X
High temperature	3.3.1	4.6.4.1	X		
Transient voltage	3.9.9	4.6.1.1.9	X		
Negative transient voltage	3.9.10	4.6.1.1.10	X		
Low temperature	3.3.2	4.6.4.2	X		
Vibration	3.3.3	4.6.4.5	X		
Shock	3.3.4	4.6.4.6	X		
Corrosion resistance	3.3.5	4.6.4.4	X		
Fungus resistance	3.3.6	4.6.4.3	X		
Waterproofness	3.3.7	4.6.4.7	X		X
Electromagnetic interference	3.9.11, 3.9.12	4.6.4.8	X		
Short circuit protection	3.9.15	4.6.1.1.11	X		
Generator accessory sensing	3.9.16	4.6.1.1.12	X		
Endurance	3.8	4.6.4.9	X		

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TABLE III. First article inspection.

Test	Requirement	Test specimens			
		1	2	3	4
Current limit	4.6.1.1.4	X	X	X	
Reverse current	4.6.1.1.1	X	X	X	X
Voltage regulation	4.6.1.1.2	X	X	X	X
Voltage ripple	4.6.1.1.3	X	X	X	
Field excitation	4.6.1.1.6	X	X	X	
High temperature	4.6.4.1		X	X	
Low temperature	4.6.4.2		X	X	
Endurance	4.6.4.9	X			
Vibration	4.6.4.5		X	X	
Shock	4.6.4.6		X	X	
Waterproofness	4.6.4.7		X	X	
Corrosion resistance	4.6.4.4		X		
Transient voltage	4.6.1.1.9		X	X	
Negative voltage operation	4.6.1.1.10		X	X	
Overvoltage	4.6.1.1.5	X	X	X	
Fungus resistance	4.6.4.3			X	
Electromagnetic interference	4.6.1.1.11				X
Field circuit	4.6.1.1.7	X	X	X	
Negative voltage operation	4.6.1.1.8	X	X		
Short circuit protection	4.6.1.1.11	X	X		
Generator accessory sensing	4.6.1.1.12	X	X		

4.5.1.2 Lot formation. An inspection lot shall consist of all regulators of one type and part number from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.2 Examination.

4.5.2.1 Classification of defects. For examination purposes, defects shall be classified as listed in table IV.

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TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Incorrect dimensions affecting interchangeability (see 3.13).	SIE 1/
Minor:		
201	Improper finish/color not as specified (see 3.11).	Visual
202	Nameplate not as specified (see 3.12).	Visual
203	Incorrect dimensions not affecting interchangeability (see 3.13).	SIE
204	Faulty workmanship (see 3.14).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Each regulator assembly shall be subjected to the conformance tests specified in table II, and in the order specified in table V.

TABLE V. Conformance tests.

Test	Paragraph
Voltage regulation	4.6.1.1.2.2
Current limit	4.6.1.1.4.2
Overvoltage	4.6.1.1.5.2
Field excitation	4.6.1.1.6
Negative voltage operation	4.6.1.1.8.2
Waterproofness	4.6.4.7.2

4.5.3.1 Failure. Failure of any regulator to pass any of the specified conformance tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.5.4 Methods of inspection.

4.5.4.1 Materials and construction. Conformance to 3.2, 3.5.1, 3.5.2, 3.6.1, 3.6.2, 3.9.14, 3.10.1, and 3.10.2, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving

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inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.4.2 Defects. Conformance to 3.5.3, 3.11, 3.12, 3.13, and 3.14, shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6 Conformance inspection.

4.6.1 Performance.

4.6.1.1 Electrical. For those tests in this specification in which pin E of the battery connector is grounded, pin E shall be grounded to the regulator case, and the regulator case shall be grounded solely through a 14 gauge wire (not longer than 20 feet (ft) (6.1 meters (m)) to the generator housing. For those tests in which pin E is not grounded, the regulator shall be grounded through its mounting bolts and braided ground strap.

4.6.1.1.1 Reverse current. To determine conformance to 3.9.1, the temperature stabilized regulator (see 6.3.1) shall be connected in the test circuit as shown on figure 4 with the generator operating at sufficient speed to deliver charging current to the batteries. The generator speed shall then be gradually reduced to zero. The maximum value of the reverse current shall be observed on the milliammeter; five test cycles shall be recorded. This test shall be performed with pin E of the regulator battery connector both grounded and ungrounded.

4.6.1.1.2 Voltage regulation.

4.6.1.1.2.1 First article. To determine conformance to 3.9.2, the temperature stabilized regulator (see 6.3.1) shall be connected in the test circuit. During the test, the resistance load shall be applied and removed suddenly. Change in load shall be accomplished in 10 ms. Tripping of the overvoltage reset due to quick reaction shall constitute failure. Observation shall be made of the output voltage under the following conditions:

PIN E BATTERY CONNECTOR UNGROUNDED		PIN E BATTERY CONNECTOR GROUNDED	
Generator (rpm)	Load (A)	Generator (rpm)	Load (A)
a. 2400 ± 50	25	a. 3000 ±50	25
2400 ± 50	100	3000 ±50	150
2400 ± 50	350		
b. 5500 ±110	same sequence as a.		
c. 8000 ±160	same sequence as a.		

Regulated voltage shall be recorded on suitable data recording tape.

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4.6.1.1.2.2 Conformance. To determine conformance to 3.9.2, the generator shall be operated at 5500 rpm with a 350 A load for 5 minutes. The regulated voltage shall then be recorded with the unstabilized regulator at an ambient temperature of 80 ±15°F (26 ±8°C) under the following conditions:

PIN E BATTERY CONNECTOR UNGROUNDED		PIN E BATTERY CONNECTOR GROUNDED	
Generator (rpm)	Load (A)	Generator (rpm)	Load (A)
a. 2400 ±50	Battery load only	a. 3000 ±50	Battery load only (current less than or equal to (≤) 150 A)
b. 2400 ±50	350 A		
c. 8000 ±160	Battery load only	b. 3000 ±50	150 A
d. 8000 ±160	350 A		

4.6.1.1.3 Voltage ripple. To determine conformance to 3.9.3, using the test procedure specified in 4.6.1.1.2, the output voltage waveform at the regulator output shall be measured with an oscilloscope with the circuit connected as shown on figure 5. Repeat the procedure with the batteries disconnected.

4.6.1.1.4 Current limit. To determine conformance to 3.9.4, the regulator-generator system shall be tested as follows. The regulator-generator system shall be operated with no batteries connected. Where applicable, the tests shall be performed under static circuit conditions using an average reading meter for voltage and current readings. For transient circuit conditions, appropriate instrumentation shall be used to verify meeting the current limit requirements of figure 3.

4.6.1.1.4.1 First article. Conformance shall be as follows:

a. To determine conformance to 3.9.4a and 3.9.4b, pin E of the regulator battery terminal shall be grounded, and the regulator shall be tested at an ambient temperature of -65 ±5°F (-53 ±3°C), 80 ±15°F (27 ±9°C), and 220 ±5°F (104 ±3°C) with a generator speed of 3000 rpm while resistive loads in 0.5 ohm ±5 percent increments are consecutively applied, beginning with no load and ending when a net load of 0.07 ohm ±5 percent is attained. These eight data points are to be plotted on a voltage-current (V-I) curve verifying conformance to figure 9.
b. To determine conformance to 3.9.4c and 3.9.4d, pin E of the regulator battery terminal shall be ungrounded, and the regulator shall be tested at an ambient temperature of -65 ±5°F, 80 ±15°F, and 220 ±5°F with a generator speed of 5500 rpm while resistive loads in 0.25 ohm ±5 percent increments are consecutively applied, beginning with no load and ending when a net load of

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0.028 ohm ±5 percent is attained. These ten data points are to be plotted on a V-I curve verifying conformance to figure 9.

c. To determine conformance to 3.9.4e, f, and g, pin E of the regulator battery terminal shall be grounded, and the regulator shall be tested at an ambient temperature of -65 ±5°F, 80 ±15°F, and 220 ±5°F with a generator speed of 3000 rpm and no load applied. A resistive load of 0.07 ohm ±5 percent shall be suddenly applied and removed and the transient data recorded. This test shall then be repeated, with pin E ungrounded, at a generator speed of 5500 rpm using a 0.028 ohm ±5 percent resistive load.

4.6.1.1.4.2 Conformance. Conformance shall be as follows:

a. To determine conformance to 3.9.4a, pin E of the regulator battery terminal shall be grounded, and the regulator shall be tested at an ambient temperature of 80 ±15°F with a generator speed of 3000 rpm with a resistive load of 0.07 ohm ±5 percent applied.
b. To determine conformance to 3.9.4b, pin E of the regulator battery terminal shall be grounded, and the regulator shall be tested at an ambient temperature of 80 ±15°F with a generator speed of 3000 rpm with a resistive load of 0.19 ohm ±5 percent applied.
c. To determine conformance to 3.9.4c, pin E of the regulator battery terminal shall be ungrounded, and the regulator shall be tested at an ambient temperature of 80 ±15°F with a generator speed of 5500 rpm with a resistive load of 0.028 ohm ±5 percent applied.
d. To determine conformance to 3.9.4d, pin E of the regulator battery terminal shall be ungrounded, and the regulator shall be tested at an ambient temperature of 80 ±15°F with a generator speed of 5500 rpm with a resistive load of 0.08 ohm ±5 percent applied.

4.6.1.1.5 Overvoltage.

4.6.1.1.5.1 First article and control. To determine conformance to 3.9.5, the regulator shall be connected to a dc power supply. The voltage shall be increased from 28 V by application of a step voltage of approximately 6 V. The voltage level and reaction time shall be measured and recorded. This test shall be performed with pin E of the battery connector both grounded and ungrounded.

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4.6.1.1.5.2 Conformance. To determine conformance to 3.9.5, the unstabilized regulator shall be disconnected from the generator and a dc voltage of 32 V applied to the regulator, the overvoltage protection circuit shall not activate. With 34 V applied to the regulator, the overvoltage protection circuit shall activate. This shall be performed with pin E of the battery connector both grounded and ungrounded.

4.6.1.1.6 Field excitation. To determine conformance to 3.9.6, the regulator shall be connected in the circuit of figure 4 with batteries disconnected, S1 open and the generator operating at a constant speed. Momentarily connect an external power supply with its polarity reversed to the generator field. Repeat this momentary connection until the residual voltage is less than 0.7 V. Record the residual voltage, close S1 and verify that the regulator excites and bootstraps the generator to its nominal output. This test shall be performed with pin E of the battery connector both grounded and ungrounded.

4.6.1.1.7 Field circuit. To determine conformance to 3.9.7, the regulator shall be tested as follows:

a. With pin E of the battery connector ungrounded, and with the generator operating at 2400 rpm in the circuit on figure 4 with a 30 A load, switch S1 shall be opened and closed five times. The load shall be increased to 300 A and the procedure repeated.
b. Procedure (a) shall be repeated with the generator operating at 8000 rpm.
c. Procedure (a) shall be repeated again with the generator operating at 3000 rpm and with pin E of the battery connector grounded.

Upon completion of this test, the regulator shall pass the test of 4.6.1.1.2.

4.6.1.1.8 Negative voltage operation.

4.6.1.1.8.1 First article. To determine conformance to 3.9.8, the regulator shall be operated until temperature stabilization. Then, it shall be subjected to the following test procedure using the test circuit shown on figure 6. (Batteries shall be fully charged and in satisfactory condition (see 6.3.2)).

a. Arrange the test as shown on the circuit diagram on figure 6 with S1, S2, S3, and S4 open.
b. Close S2 for 1 to 5 seconds; then open S2.
c. Close S3 with the generator at 2400 ±50 rpm. Observe the generator output, which should be less than 200 A. If the generator output indicates 200 A or more, open S3 and discontinue the test. If the generator output does not exceed 200 A, close S1 and test the regulator in accordance with 4.6.1.1.2.2, 4.6.1.1.4, and 4.6.1.1.5.2.

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d. Open S1, S2, and S3. Lock the generator to prevent rotation (rotation less than 10 rpm is permissible). Close S3, and reverse the polarity of V2 for 10 seconds; then return the polarity to normal. Repeat step (c), and determine that the output is in accordance with 4.6.1.1.2.2, 4.6.1.1.4, and 4.6.1.1.5.2.

NOTE: Steps (b) and (c) test for ability to withstand reverse generator operation. Step (d) tests the ability to withstand reverse slave battery operation.

Upon completion of this test, the regulator shall pass the tests of 4.6.1.1.2.2, 4.6.1.1.4, and 4.6.1.1.5.2.

4.6.1.1.8.2 Conformance. To determine conformance to 3.9.8, the regulator shall be subjected to the following procedure using the test circuit of figure 6.

a. Close S2 from 1 to 5 seconds, then open.
b. Lock the generator to prevent rotation (rotation of less than 10 rpm is permissible). Close S3 and reverse the polarity of V2 for 10 seconds. Return polarity to normal.

4.6.1.1.9 Transient voltage. To determine conformance to 3.9.9, the regulator shall be disconnected from the test circuit shown on figure 4 and stored for 24 hours at 215 ±5°F (101 ±3°C). The test shall then be conducted in an ambient temperature of 215 ±5°F. A voltage pulse of 100 V, square wave, not more than 160 microseconds (ms) rise time, 50 ms duration, shall be applied to the regulator output terminal once every 6 seconds, equally spaced for a period of 1 hour with pin E of the battery connector ungrounded and 3 hours with pin E grounded. The test shall be repeated, except that a voltage pulse of 70 V shall be applied to the input terminal of the generator.

4.6.1.1.10 Negative transient voltage. To determine conformance to 3.9.10, the regulator shall be temperature stabilized at 215 ±5°F for a period of 1 hour before being subjected to five cycles of the following two-step procedure, using the test circuit shown on figure 7. Steps one and two shall constitute a cycle, and the cycle shall be repeated five times.

Step 1: a. Capacitor C shall be charged to a voltage of 595 ±5 V.
b. Switch S2 shall be closed (voltage is applied to the input terminal of the regulator) for a period of 6 seconds and then opened.

Step 2: a. With the generator running at a speed between 2400 and 3000 rpm, switch S3 shall be closed for 1 minute.
b. Capacitor C shall be charged to a voltage of 595 ±5 V and switch S2 shall be closed for a period of 6 seconds and then opened.
c. Open all switches.

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Step 3: Repeat steps 1 and 2 with pin E of the battery connector grounded through S1.

4.6.1.1.11 Short circuit protection. To determine conformance to 3.9.15, with the regulator connected and operating in the test circuit of figure 4, and with pin E of the battery connector grounded, connect pin "D" of the Generator connector to ground and maintain this connection for 10 seconds minimum. Remove the ground connection, depress the reset switch and repeat this procedure for pins "F" and "H". This test shall be repeated with pin E of the battery connector ungrounded. Subsequent to the test the regulator shall meet the requirements of 4.6.1.1.2.

4.6.1.1.12 Generator accessory sensing circuit To determine conformance to 3.9.16, a resistive load shall be connected to pin "H" and the regulator shall be operated in the test circuit of figure 4. With a minimum of 0.5 A load current on pin "H", the voltage on pin "H" shall be observed and recorded to be within the specified limits.

4.6.2 Interface requirement. The regulator is to conform to all interfaces of electrical and mechanical as depicted on Drawing 12333040, and associated drawings.

4.6.3 Reliability. The Government reliability engineer shall determine the reliability criteria prior to finalization of the contract (see 6.2).

4.6.4 Environmental conditions.

4.6.4.1 High temperature. To determine conformance to 3.3.1, the regulator shall be stored for 24 hours in an ambient temperature of 220 ±5°F (104 ±3°C). With this ambient temperature maintained, the regulator shall function continuously for 1 hour while the generator speed and output load are adjusted to 2400 rpm and 350 A with pin E of the battery connector ungrounded. The regulator shall then function continuously for 4 hours while the generator speed and output load are adjusted for 3000 rpm and 150 A with pin E of the battery connector grounded. While remaining in an ambient temperature of 220 ±5°F, the regulator shall be tested as specified in 4.6.1.1.1, 4.6.1.1.2.1, 4.6.1.1.5.1, 4.6.1.1.6, 4.6.1.1.7, 4.6.1.1.9, and 4.6.1.1.10. The regulator shall be returned to an ambient temperature of 80 ±5°F (26 ±3°C) and tested as specified in 4.6.1.1.1, 4.6.1.1.4, 4.6.1.1.2.2, and 4.6.1.1.5.2.

4.6.4.2 Low temperature. To determine conformance to 3.3.2, the regulator shall be stored for 24 hours in an ambient temperature of -80 ±5°F (-62 ±3°C). After stabilizing for 4 hours at -65 ±5°F (-54 ±3°C), the regulator shall function continuously for 1 hour while the generator speed and output load are adjusted to 2400 rpm and 400 A with pin E of the battery connector ungrounded. The regulator shall then function continuously for 1 hour while the generator speed and output load are adjusted to 3000 rpm and 150 A with pin E of the battery

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connector grounded. While remaining in the ambient temperature of -65 ±5°F (-54 ±3°C), the regulator shall be tested as specified in 4.6.1.1.1, 4.6.1.1.2.1, 4.6.1.1.5.1, 4.6.1.1.6, 4.6.1.1.7, 4.6.1.1.9, and 4.6.1.1.10. The regulator shall then be returned to an ambient temperature of 80 to ±5°F (26 ±3°C) and tested as specified in 4.6.1.1.1, 4.6.1.1.2.2, 4.6.1.1.4.2, and 4.6.1.1.5.2.

4.6.4.3 Fungus resistance. To determine conformance to 3.3.6, documentation shall be furnished providing evidence that exposed materials of the regulator are non-nutrient to fungi or have been treated to resist fungi detrimental to the performance of the regulator. In the absence of such documentation, the regulator shall be subjected to the fungus resistance test in accordance with MIL-F-13927 for class 2 components, except that the period of exposure shall be 90 days. Sealed and internal components shall not be exposed. The regulator shall subsequently be tested in accordance with 4.6.1.1.2.1.

4.6.4.4 Corrosion resistance. To determine conformance to 3.3.5, the regulator shall be subjected to the corrosion resistance test in accordance with ASTM D522 Mandrel Bend Test and GM 9540P, Method B, 120 cycles or until one of the items fail with defects such as extensive corrosion at scribe or significant penetration of base metal. The regulator shall subsequently be tested in accordance with 4.6.1.1.2.1.

4.6.4.5 Vibration. To determine conformance to 3.3.3, the regulator, mounted as it will be in use, shall be subjected to tests as specified in MIL-STD-202, method 204D, Test condition A, except that a constant force of 1.5 gravity units shall be applied between 5.5 and 22 Hz. The sweep frequency shall be varied logarithmically. The sweep shall be a simple harmonic motion, and its amplitude shall have a tolerance of ±10 percent. Upon completion of this test, the regulator shall pass the tests of 4.6.1.1.1, 4.6.1.1.2.1, 4.6.1.1.4, and 4.6.1.1.5.1.

4.6.4.6 Shock. To determine conformance to 3.3.4, the regulator shall be mounted as it will be in use and subjected to tests in accordance with MIL-STD-202, method 213B, Test condition G. The regulator shall subsequently be tested in accordance with 4.6.1.1.1, 4.6.1.1.2.1, 4.6.1.1.4, and 4.6.1.1.5.1.

4.6.4.7 Waterproofness.

4.6.4.7.1 First article and control. To determine conformance to 3.3.7, the regulator shall be subjected to the waterproofness test in accordance with MIL-STD-1184, type II, class 2, except 10 pounds per square inch (psi) (0.07 pascals (Pa) pressure and 10 psi vacuum shall be used. The regulator, while submerged, shall be operable and tested in accordance with 4.6.1.1.2.2 during the pressure and vacuum test phases. The regulator shall subsequently be tested in accordance with 4.6.1.1.2.2 and 4.6.1.1.4.

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4.6.4.7.2 Conformance. To determine conformance to 3.3.7, the nonoperating regulator shall be subjected to the conformance test inspection as specified in MIL-STD-1184, type II, class 2. During submersion, the regulator shall be subjected to 10 pounds per square gauge (psig) for 5 minutes.

4.6.4.8 Electromagnetic interference. To determine conformance to 3.9.11 and not exceed the limits of MIL-STD-461, the regulator shall be laboratory tested in accordance with MIL-STD-462 while operating a balanced 300 A split load in a circuit as shown in figure 5. The radiated emissions are to be measured at a distance of 3 ft (0.91 m).

4.6.4.9 Endurance. To determine conformance to 3.8 the regulator shall be temperature stabilized (see 6.3.1). With pin E of the battery connector grounded, the regulator shall first be cycled at 18 cycles per hour for 1000 hours. A cycle shall consist of generator operation, at a speed of 3000 rpm for 25 seconds each at battery load, 50 A, 100 A, 150 A, and 195 $\pm$15 A, followed by a reduction to 150 A, 100 A, 50 A and then battery load. Pin E of the regulator battery connector shall then be ungrounded and the system cycled at 18 cycles per hour for 100 hours. The 100 hours shall be divided into 25 hours with generator operating at 2400 rpm, 25 hours at 4000 rpm, 25 hours at 6000 rpm, and 25 hours at 8000 rpm. A cycle shall consist of generator operation for 25 seconds each at battery load, 100 A, 200 A, 300 A, and 400 A, followed by a reduction to 300 A, 200 A, 100 A, and then battery load. Subsequently, the regulator shall be tested in accordance with 4.6.1.1.1, 4.6.1.1.2.1, 4.6.1.1.3, 4.6.1.1.4.1, 4.6.1.1.5.1, and 4.6.4.7.1

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The regulators covered by this specification are intended for use in 28 Vdc nominal electrical systems equipped with charging generators. The regulators are intended for use on tactical and combat military vehicles.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article samples are required (see 3.1 and 4.4).
d. Quantitative Reliability numbers for regulators (see 3.7 and 4.6.3).
e. If finish should be other than as specified (see 3.11).
f. Drawing title, number, and date (see 3.12).
g. If inspection conditions should be other than as specified (see 4.2).
h. If responsibility for inspection equipment should be other than as specified (see 4.3).
i. If first article sample size should be other than as specified (see 4.4).
j. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Temperature stabilized regulator. A temperature stabilized regulator is defined as a regulator which has been operated for 30 minutes with the generator speed adjusted to 3000 rpm, the load adjusted to 175 A, and the temperature at 80 ±15°F (26 ±8°C) with pin E of the battery connector grounded (see 3.8, 4.6.1.1.1, 4.6.1.1.2.1, and 4.6.4.9).

6.3.2 Fully charged battery. For the purposes of the test procedures in this specification, a fully charged battery is defined as a battery having a current consumption of not more than 4 A at 28 V at a room temperature of 80 ±15°F (26 ±8°C) (see 4.2.1 and 4.6.1.1.8.1).

6.3.3 Load current. Load current for purposes of this specification should be considered as that current from generator and regulator to batteries and resistive load (see figure 4).

6.3.4 Radioactive material. Radioactive material is defined by Title 10, Code of Federal Regulations, Part 40, and other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for this item (see 3.2.1).

6.4 Disposition of test assemblies. Assemblies undergoing destructive tests should not be used and should be indelibly marked "DO NOT USE", and purposely scrapped.

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6.5 Subject term (key word) listing.

Series-parallel
Temperature stabilized
Voltage overshoot
Voltage undershoot.

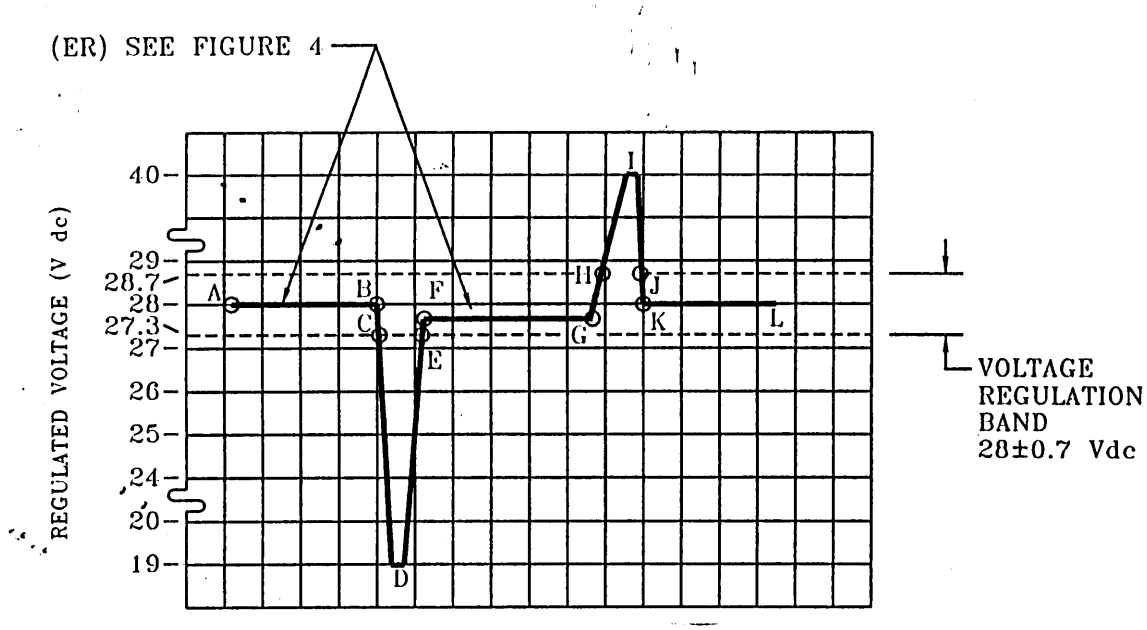
6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONTRACT DESIGNATION	GENERATOR CONNECTOR	BATTERY CONNECTION
A	(+) Generator main engine	Battery connection
B		
C		
D	Generator field	
E		Low current limit ground connection
F	Generator field	
G	(+) Generator main engine	Battery connection
H	Generator accessory sensing circuit	

FIGURE 1. Contact designation data.

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(A-B) - BATTERY LOAD ONLY PRIOR TO LOAD APPLICATION

(B) - POINT OF LOAD APPLICATION

(C-E) - VOLTAGE UNDERSHOOT RECOVERY TIME

(D) - UNDERSHOOT VOLTAGE

(F-G) - REGULATED VOLTAGE (ER) DURING APPLIED LOAD (STEADY RATE)

(G) - POINT OF LOADS DISCONNECTION

(H-J) - VOLTAGE OVERSHOOT RECOVERY TIME

(I) - OVERSHOOT VOLTAGE

(K-L) - BATTERY VOLTAGE AFTER LOAD DISCONNECT

(C-E) - MAXIMUM TIME LIMIT - 0.125 SECOND

(H-J) - MAXIMUM TIME LIMIT - 0.300 SECOND

FIGURE 2. Voltage regulation limits.

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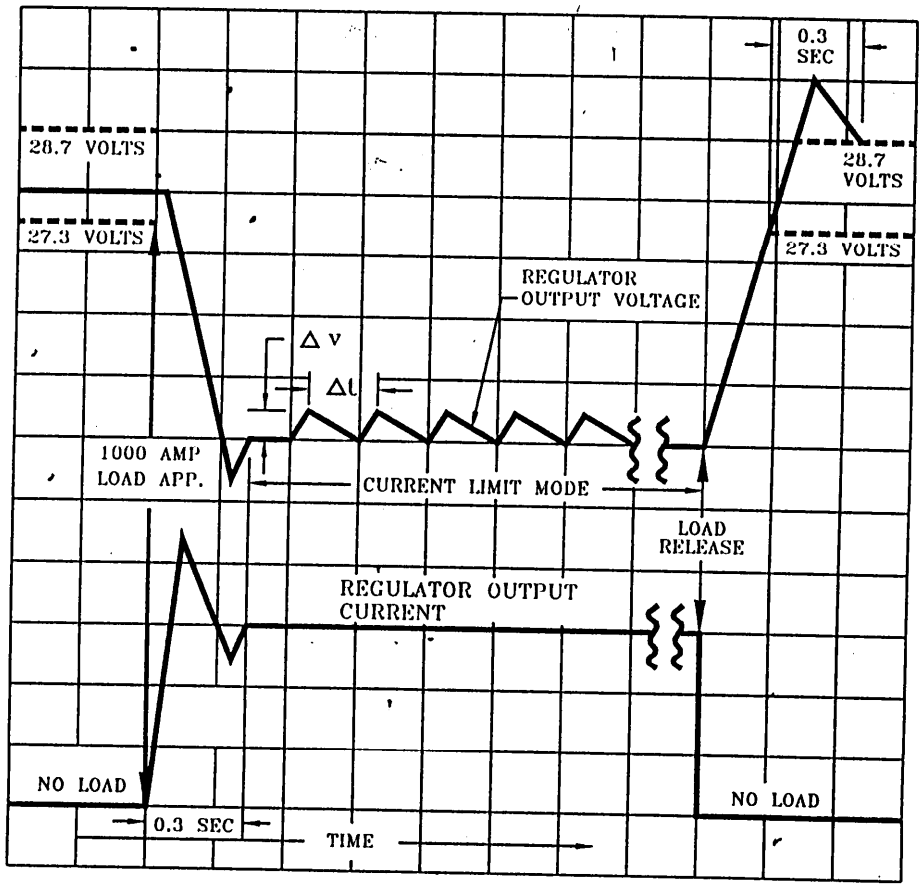


FIGURE 3. Current and voltage recovery time.

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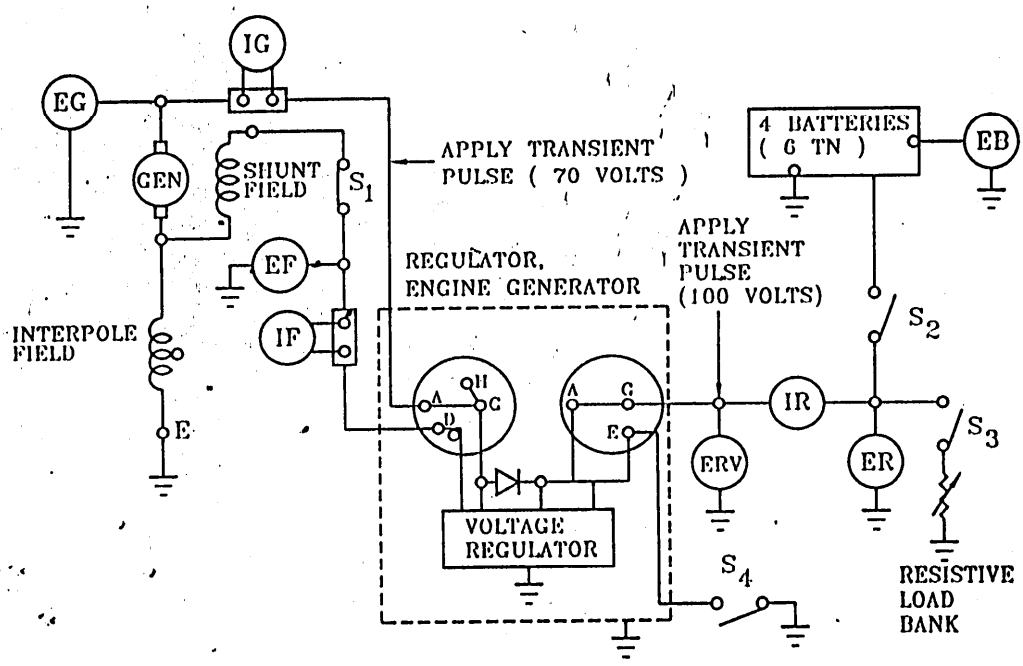


FIGURE 5. Electromagnetic interference tests

TEST DATA REQUIRED

IG - GENERATOR (GEN) AMPS
EB - BATTERY VOLTS
EG - GEN VOLTS
EF - GEN FIELD VOLTS
RL - RESISTIVE LOAD BANK
IR - REVERSE CURRENT
S_1 - SWITCH USED DURING FORDING MS39061-1
ER - REGULATED VOLTAGE
ERV - RIPPLE VOLTAGE
IF - GEN FIELD AMPS

FIGURE 4. Test circuits.

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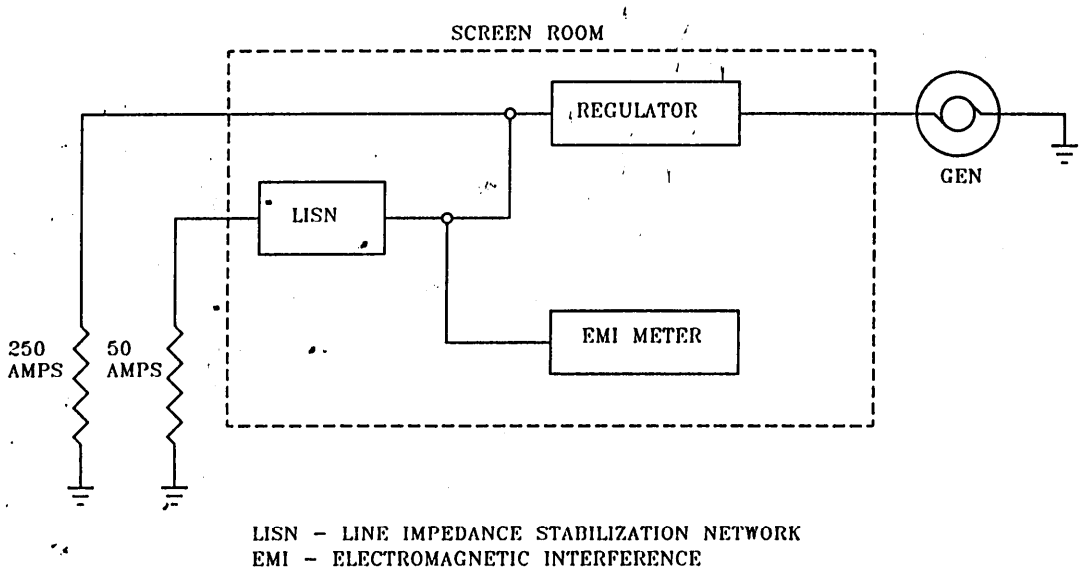


FIGURE 5. Electromagnetic interference tests.

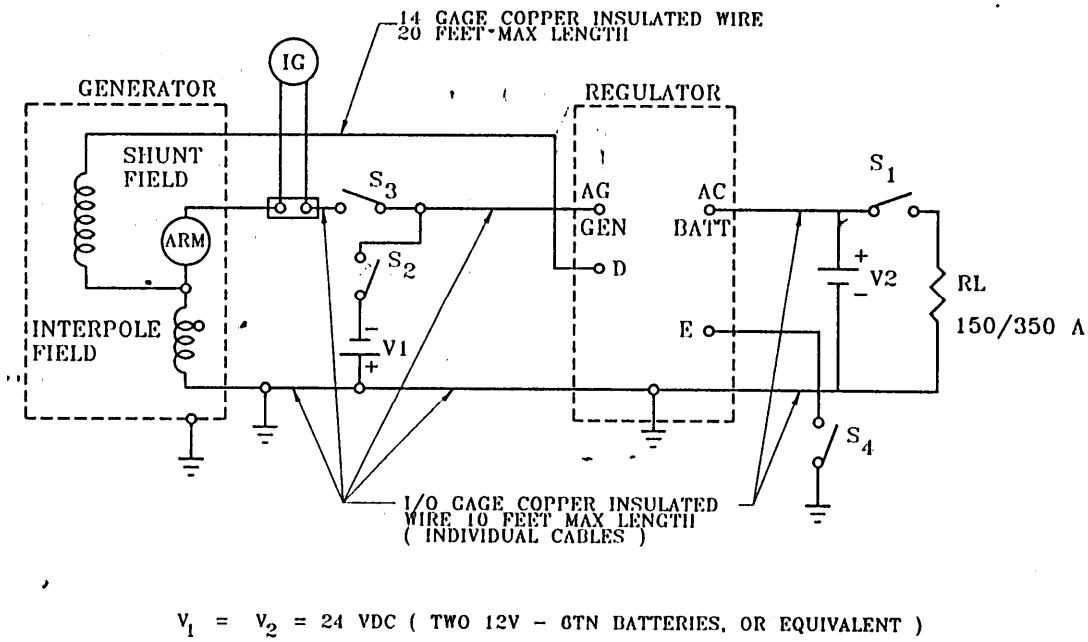


FIGURE 6. Negative voltage circuit.

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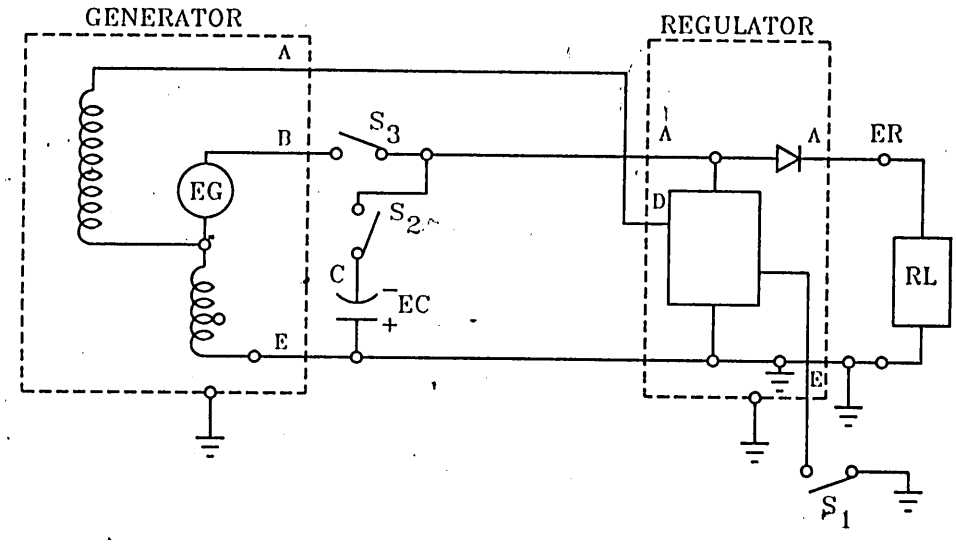


FIGURE 7. Test circuit.

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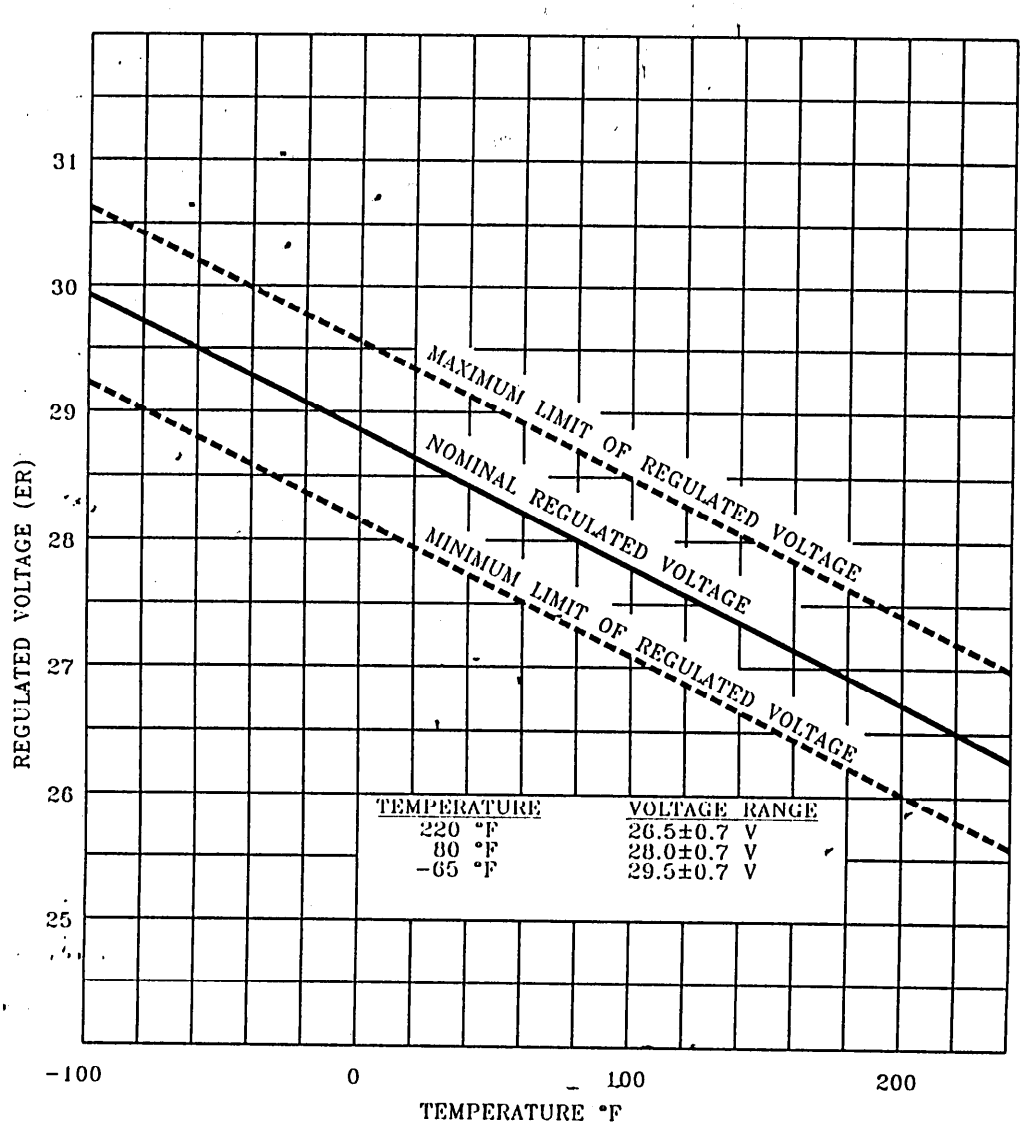


FIGURE 8. Voltage regulator temperature compensation.

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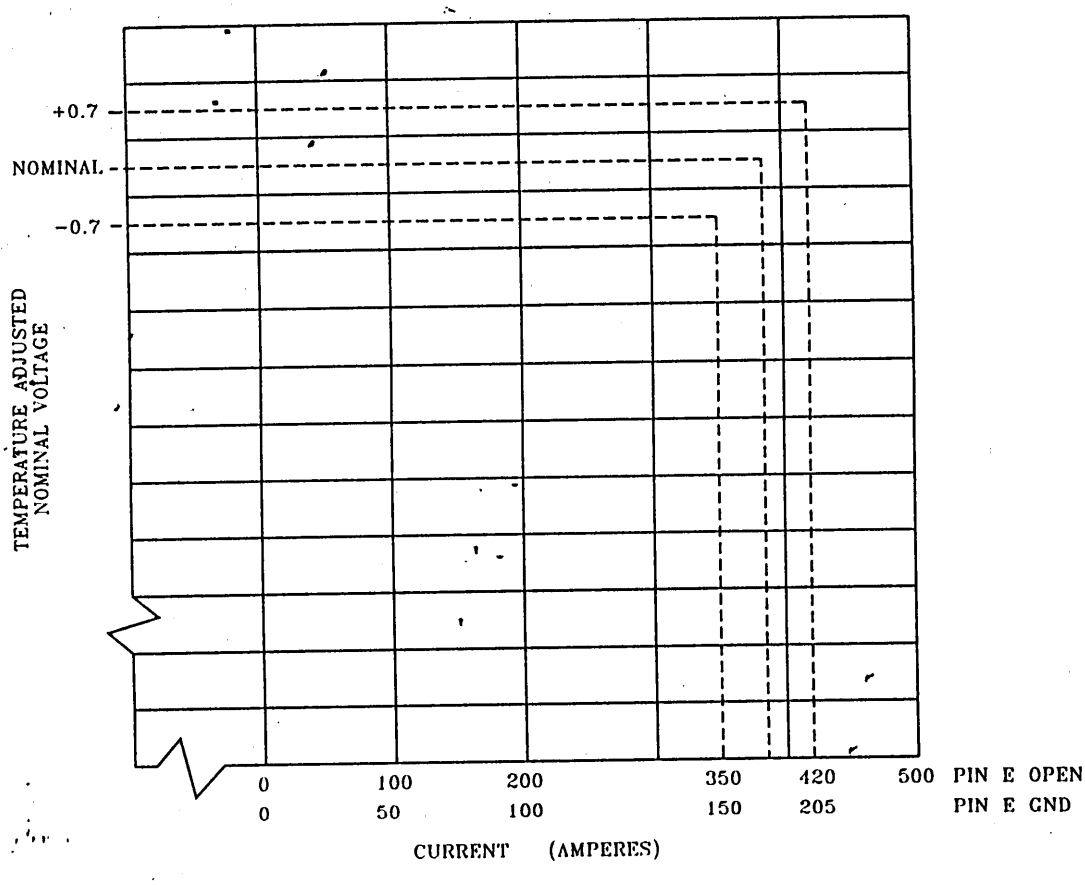


FIGURE 9. Universal current limit Performance-requirements.

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Preparing Activity:
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(Project 2920-0456)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-62528A(AT)	2. DOCUMENT DATE (YYMMDD) 960726

3. DOCUMENT TITLE Regulator, Engine Generator: Solid State 28 Volt, Dual current Limit, Direct Current

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial: (810) 574-8745 (2) AUTOVON: 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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